

After our last meeting, I realized I had been answering at the wrong level of abstraction. You asked for high-level directions, but what I gave you were implementation decisions — things like using color to encode sentiment or size to encode frequency. Those are design choices, not research questions. What I want to do this time is step back, articulate what questions I actually want to explore and what I want to reveal, and only then talk about how to visualize it.

Overall line of thinking

My starting point is a phrase in the project brief: moving away from geographical maps. Most people would read this as "use a more abstract visual form." But I think there's a deeper implication: geographical maps assume that the meaning of a place name is determined by its location. In literature, though, the meaning of a place name is determined by its narrative context — when it appears, what other place names it appears alongside, and which author chose it. None of these can be expressed on a map. So the core question I want to explore is: **can place names in literary text be visualized as an independent narrative structure, rather than simply as another rendering of geographic information?**

From this question, I'm drawn to four specific directions.

Direction 1: The Author's Literary-Geographic Fingerprint

The research question is: given only the distribution of place names in a text, can we identify the author without knowing their name?

The intuition I want to challenge is that place names are just background filler. If place name selection is a narrative decision, then different authors' patterns of selection should be systematically different — just as vocabulary choice or syntactic rhythm constitutes part of a writer's style. Stevenson's Edinburgh clusters around the historical buildings and narrow closes of the Old Town, steeped in darkness — entirely consistent with his Gothic sensibility. Mary Shelley's spatial choices are entirely different. This difference is not accidental, and it should be possible to visualize it.

The visualization goal is to construct a "spatial signature" — a unique abstract visual form for each author, entirely independent of the map, so that a viewer can distinguish one author's spatial language from another's at a glance.

This direction has direct support in the literature. Stange & Dörk's Novel City Maps already demonstrates the concept of a novel's "spatial fingerprint" through their street map view, where brightness encodes frequency to give each novel a distinct visual form on the map. I want to push this idea further by abandoning geographic coordinates entirely and defining the signature purely through narrative relationships.

Feasibility: The existing data's `author` field combined with `location_id` allows direct computation of each author's place name distribution vector. With 548 documents spanning multiple authors, the dataset is large enough to support this direction.

Direction 2: Narrative Topology — Place Co-occurrence Relationships

The research question is: which places are repeatedly linked by the same narrative context, and does this "narrative proximity" diverge from geographic proximity?

I want to construct a network entirely defined by textual relationships: nodes are places, and edges represent how frequently two places co-occur within the same passage. This network has no reference to the real map. It might reveal counter-intuitive structures — two geographically distant places are narratively tightly bound, while two geographically adjacent places never appear together. The divergence between "narrative distance" and "geographic distance" is itself the finding: it reflects how literary imagination reorganizes urban space.

This direction maps directly onto the London Tube Map reference in the project brief — the Tube Map is also an abstract topology that replaces real geography with functional connection. Stange & Dörk's transit map view operates on exactly this logic, using a metro-map aesthetic to reveal co-occurrence structures between places in a story.

Feasibility: The `sentence_id` field in `api_locationmention.csv` allows direct identification of whether two places appear in the same sentence; `page_id` can define a looser "same scene" co-occurrence. The co-occurrence matrix can be computed directly from this table with no additional data required.

Direction 3: The Limits of the Data — Narrative Weight is Not Uniform

This direction addresses a more fundamental issue: before deciding what to visualize, we need to ask what the data is actually measuring.

The existing database records the fact of a place name's occurrence, but not the narrative quality of that occurrence. A place name appearing in the climactic monologue of a chapter carries entirely different meaning from the same place name appearing in passing background description. If we only look at frequency, we get a flattened, distorted picture of the spatial landscape.

I think at least three dimensions can distinguish narrative weight: first, the position of the place name in the text — is it at the opening of a chapter, a key moment of scene-setting, or in transitional description? Second, does it appear in direct character dialogue or in narrative prose? Third, what is the emotional density of the surrounding context?

These features can all be extracted from the source text, but require augmenting the existing database. This directly responds to the project brief's explicit statement that "the extraction of additional meta data is required as part of this project" — this is not merely a technical task, but the prerequisite for building visualizations on a reliable analytical foundation.

Feasibility: The existing data already includes `start_word` and `end_word` fields recording the precise word-level position of each place name in the text, meaning the original text's

word-level information is traceable. Extracting contextual features on this basis is technically feasible.

Direction 4: The Geography of Literary Silence

This direction is the logical endpoint of the whole framework: if place names appearing is data, then place names not appearing is also data.

Which significant spaces of Edinburgh — working-class neighborhoods, industrial ports, factories — never or rarely entered the literary imagination? These silences are not random omissions but traces of the systematic bias of literature as a cultural practice. Making silence visible is the central task.

Operationally, this requires a dialogue between two reference systems: the "written Edinburgh" produced by the literary data, and the "real Edinburgh" produced by historical geographic data. The difference between the two is literature's blind spot. The visualization challenge is making "absence" visually perceptible — for instance, encoding literary attention through density or saturation, so that persistently dark zones on the resulting map are the actual finding.

This research question has a clear position in digital humanities scholarship under the concept of "the geography of literary silence." Loxley et al. in the Palimpsest paper also raise the issue of literature's selective coverage of urban space, noting how the choice and frequency of place names shaped both the text mining process and the resulting visualizations.

Feasibility: The `api_location.csv` table includes `lat/lon` coordinates and a `p_type` field (place type, such as suburb or neighbourhood) for all 2,135 locations. These can be compared against Edinburgh's historical geographic data to identify which area types are systematically absent from the literary record.

第一层：整个文档的出发点

你从项目简介里的一句话开始：**"moving away from geographical maps"**。

大多数人读到这句话，会想：好，那就换个更抽象的画法——比如用圆圈代替地图标记，用线条代替街道。

但你的思路更深一层：**为什么要离开地理地图？**

你的回答是：因为地理地图有一个隐含假设——一个地名的意义由它的位置决定。"卡尔顿山"在地图上就是那个坐标点。

但在文学里，这个假设是错的。"卡尔顿山"出现在斯蒂文森的小说里，它的意义不是那个经纬度，而是：

- 它在第几章出现
- 它前后出现了哪些其他地名
- 是斯蒂文森选择了它，而不是别的作者

这三件事没有一件能在地图上表达。

所以你的核心问题变成了：能不能把文学地名当作一种独立的叙事结构来可视化，而不是把它当作地理信息的另一种渲染？

这就是整份文档的地基。四个方向都是从这个问题的地基里长出来的。

第二层：四个方向各自在问什么

方向 1——作者的空间指纹

核心直觉： 作家选择哪些地名，本身就是一种风格。

就像你能从一个作家的用词习惯认出他，你也应该能从他选择的地名认出他。

斯蒂文森写的爱丁堡，集中在旧城区的历史建筑和狭窄巷弄，充满黑暗感——这和他的哥特风格完全一致。这不是巧合。

它想证明的事： 地名选择不是背景填充，是叙事决策，是作者风格的一部分。

可视化目标： 为每个作者构建一个"空间签名"——一个抽象的视觉图形，完全不依赖地图坐标，只用地名之间的关系来定义。看一眼就能区分斯蒂文森和麦卡尔·史密斯。

和已有研究的关系： Stange & Dörk 的"街道地图视图"已经做了类似的事——用亮度编码频率，让每部小说在地图上有独特的视觉形态。你想更进一步：**把地图坐标也扔掉**，只用叙事关系定义签名。

方向 2——叙事拓扑网络

核心直觉： 两个地名经常在同一段话里同时出现，说明它们在叙事上是"邻居"，但它们在地理上不一定相邻。

它想揭示的事： 文学想象对城市空间的重组。

可能会发现：

- 两个地理上很远的地方，叙事上高度绑定（比如某作家每次提到 Leith 就会提到 Holyrood）
- 两个地理上相邻的地方，叙事上几乎从不同时出现

这个"叙事距离"和"地理距离"之间的偏差，本身就是研究发现。

可视化目标： 构建一张纯粹由文本关系定义的网络图。节点是地名，边代表共现频率。这张图和真实地图长得完全不同——它展示的是文学如何重新组织城市空间。

和已有研究的关系： 这正是地铁图隐喻的本质——伦敦地铁图也是把真实地理替换成功能连接关系。Stange & Dörk 的"叙事视图"也是这个逻辑。你在爱丁堡语料上做同样的事。

方向 3——叙事权重不均匀

核心直觉： 现有数据库只记录了地名"出现了"，但没记录"出现得重不重要"。

同一个地名，出现在章节开头的场景铺垫里，和出现在高潮段落的关键独白里，意义完全不同。但数据库把它们当成一样的记录。

你想揭示的问题： 只看频率，会得到一张失真的空间图景。

你认为可以区分叙事权重的三个维度：

1. 位置——地名在文本中的位置（章节开头？过渡段？）
2. 语境——出现在人物对话里还是叙事散文里
3. 情感密度——周围文本的情感强度

关键问题： 这个方向需要原始文本才能做，但你现在的数据库里没有原文。这是一个真实的数据缺口。

它的意义： 这个方向是在说——在我们决定可视化什么之前，先要搞清楚数据在测量什么。这是一种方法论上的自我批判，学术价值很高，但执行成本也最高。

方向 4——文学沉默的地理

核心直觉： 如果地名出现是数据，那地名不出现也是数据。

爱丁堡的工人阶级社区、工业港口、工厂——这些地方在文学里几乎不存在。这不是随机遗漏，而是文学作为一种文化实践的系统性偏见的痕迹。

它想做的事： 让"缺席"变得可见。

具体操作：

- "文学爱丁堡"：你的数据里出现过的地名，按频率/密度分布
- "真实爱丁堡"：历史地理数据里的完整城市空间
- 两者相减：文学从未凝视过的地方

可视化挑战： 怎么让"没有"在视觉上被感知到——比如用密度/饱和度编码文学关注度，持续黑暗的区域就是真正的发现。

学术定位： 这个方向在数字人文领域有明确的概念位置，叫"the geography of literary silence"（文学沉默的地理）。Loxley et al.的论文也提到了文学对城市空间的选择性覆盖问题。

第三层：四个方向的逻辑关系

它们不是并列的四个独立想法，而是有内在结构的：

核心问题：地名作为叙事结构，而非地理信息

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├— 方向 1：单个作者维度——地名选择反映作者风格

├— 方向 2：地名关系维度——地名之间的叙事连接

├— 方向 3：单次出现维度——每次出现的质量不同（需要原文）

└— 方向 4：整体缺失维度——哪些地方从未被选择

方向 1 和 2 是正向的——**分析出现了什么**。方向 4 是反向的——**分析没出现什么**。方向 3 是元层面的——**质疑现有数据的测量方式**。